SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522

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Register Number	192511377

MODEL PRACTICAL EXAMINATION – SET 6

Answer all questions. Each question carries equal marks (20). *Total marks: 100*

1. Create a simple cloud software application for Online Super Market using any Cloud Service Provider to demonstrate SaaS, with the template of Customer Name, Address, Phone, Email, Items, Quantity, etc. along with the item classification form.

2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.

3. Install VirtualBox with different flavours of Linux or Windows OS on top of Windows 10 using custom installation.

4. Write a procedure to run virtual machines of different configurations and check how many virtual machines can be utilized at a particular time.

1. Create a simple cloud software application for Online Super Market using any Cloud Service Provider to demonstrate SaaS, with the template of Customer Name, Address, Phone, Email, Items, Quantity, etc. along with the item classification form.

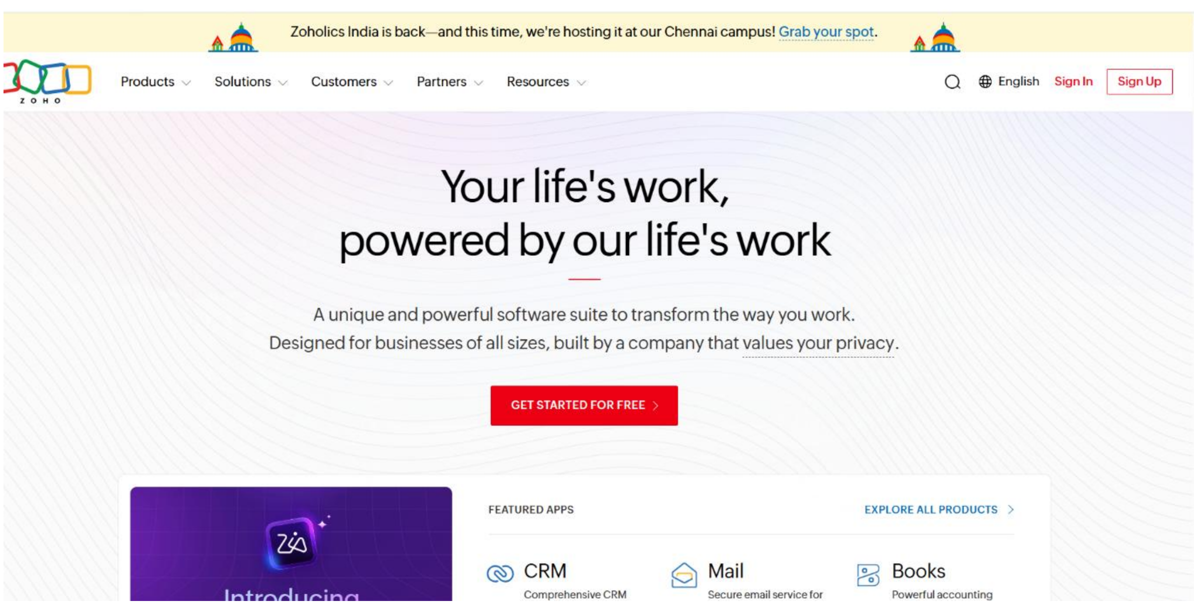
Aim:
To create a cloud-based Online Supermarket System using Zoho Creator.

Steps:

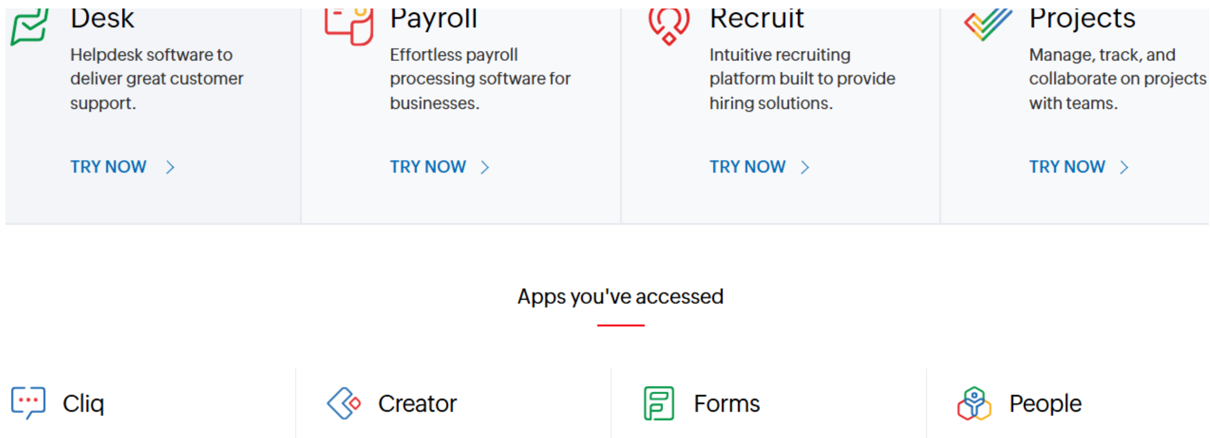
1. Create a new application.
2. Create an **Online Supermarket** form.
3. Add fields: Customer Name, Address, Phone Number, Email, Item Name, Quantity, Price, Item Classification, and Total Amount.
4. Create reports to view customer orders.
5. Enter sample records and verify the application.

Implemented:

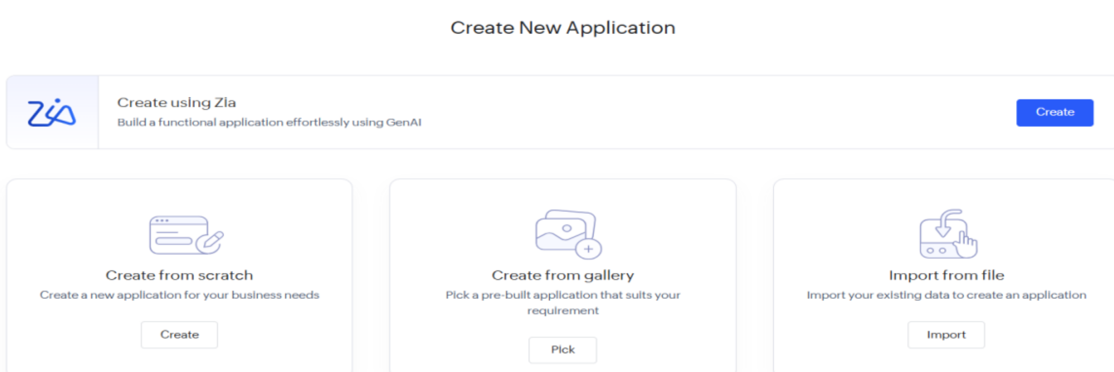
Step 1: Go to zoho.com



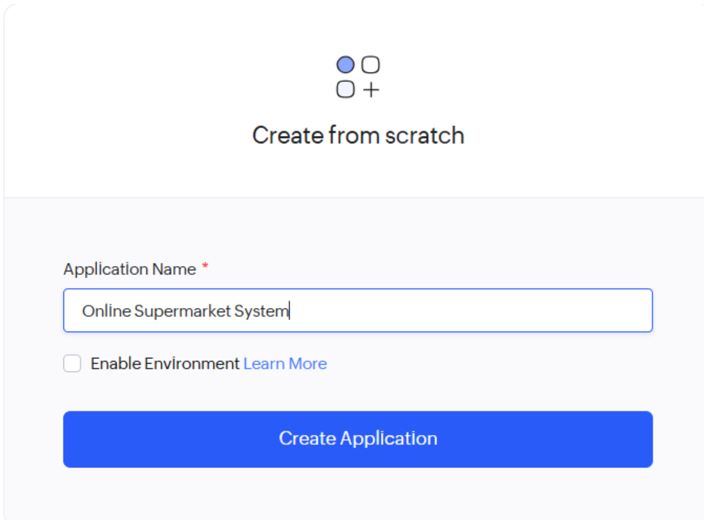
STEP 2: Login to your Zoho account and open Zoho Creator



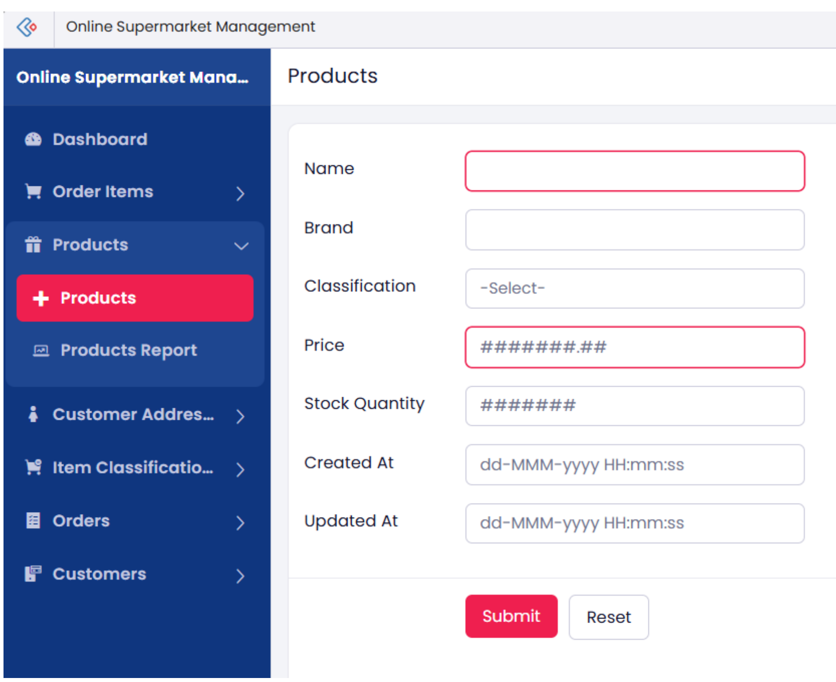
STEP 3: Click Create New Application and select Create from Scratch

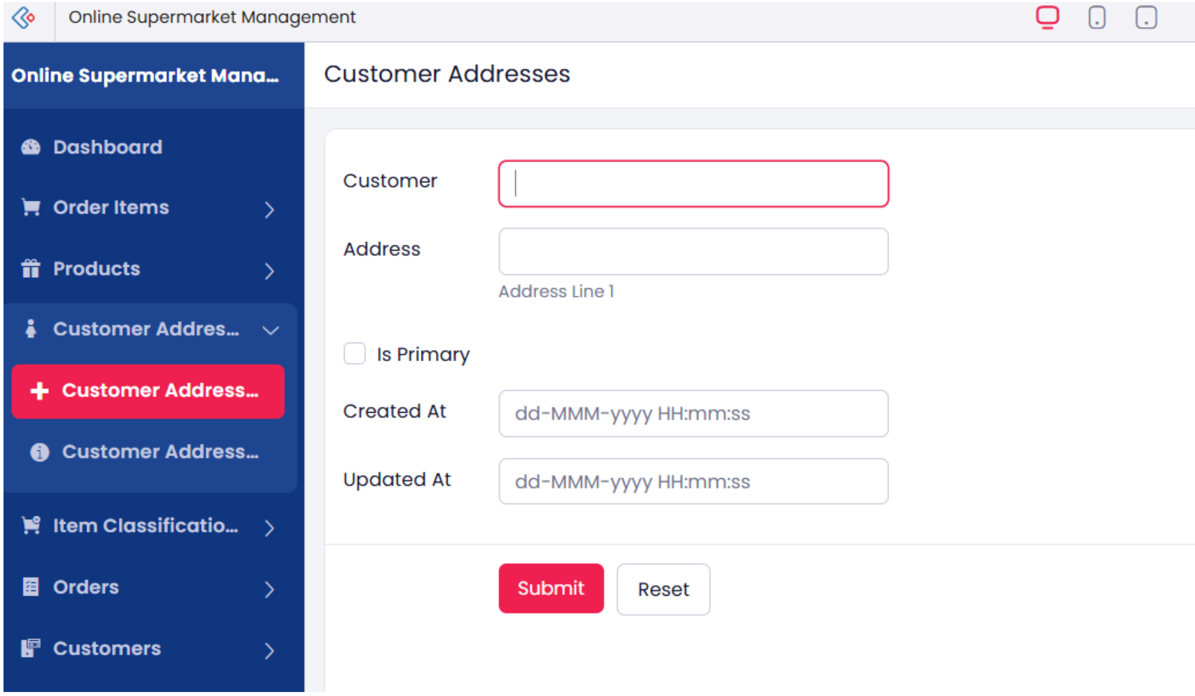


STEP 4: Enter Online Supermarket System as the application name and click Create Application.

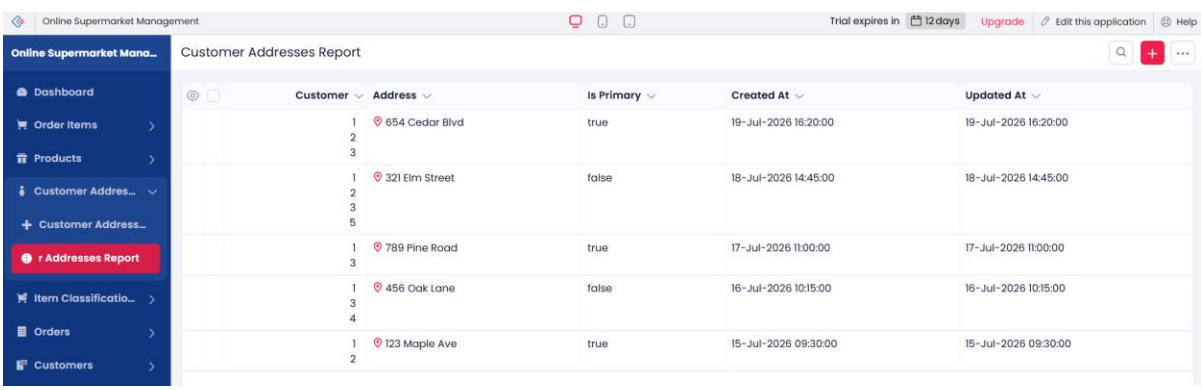


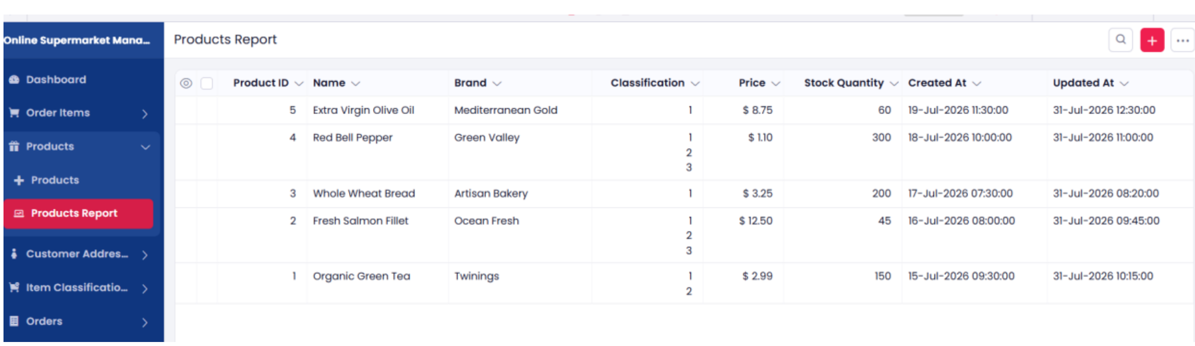
Step 5: Add fields: Customer Name, Address, Phone Number, Email, Item Name, Quantity, Price, Item Classification, and Total Amount.



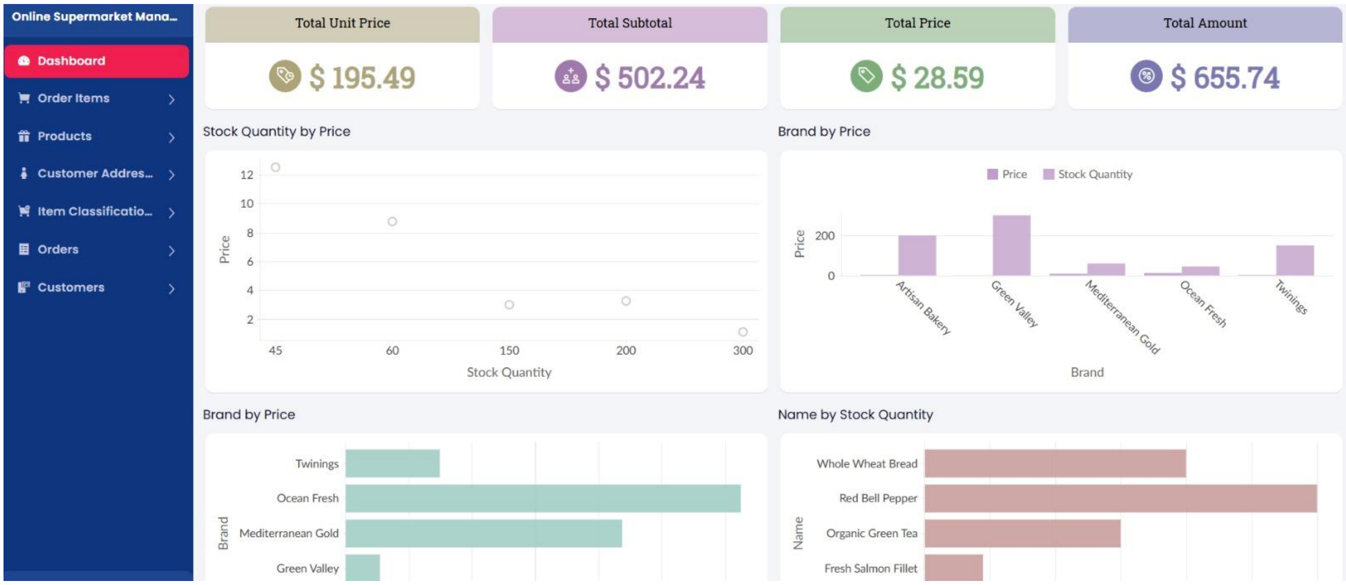


STEP 6: Save the form Online Supermarket System, enter sample booking records, and view them in the System





Step 7: The Software has Been Created



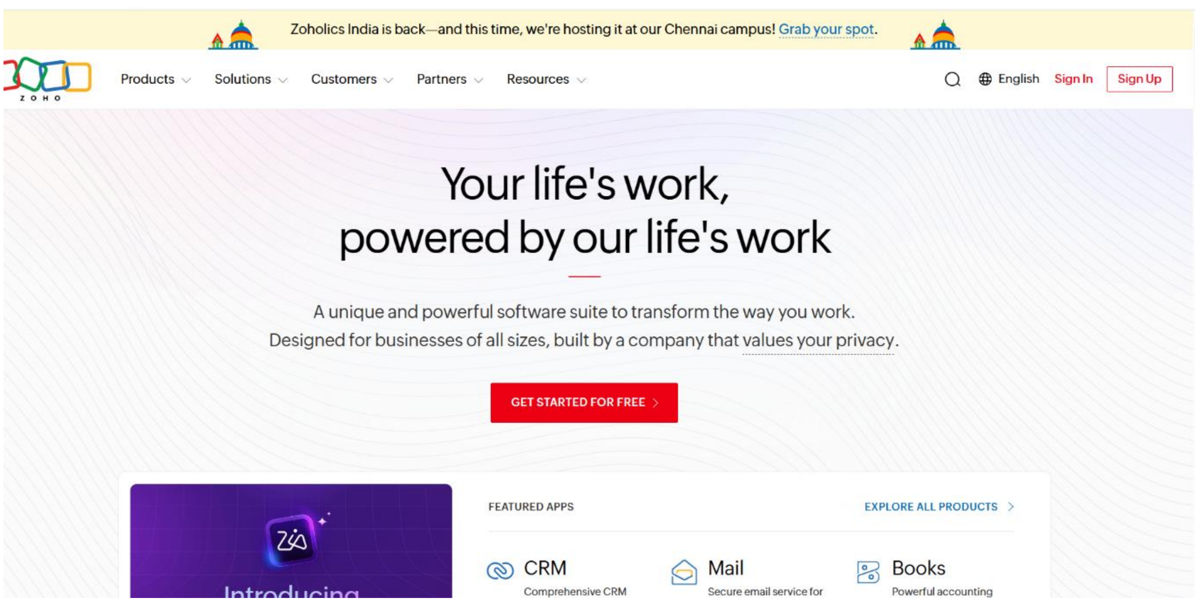
Result: The Online Super Market cloud application was successfully created with customer details, item classification, quantity, and billing-related fields. The application demonstrated the basic working of SaaS on a cloud platform.

2. . Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc

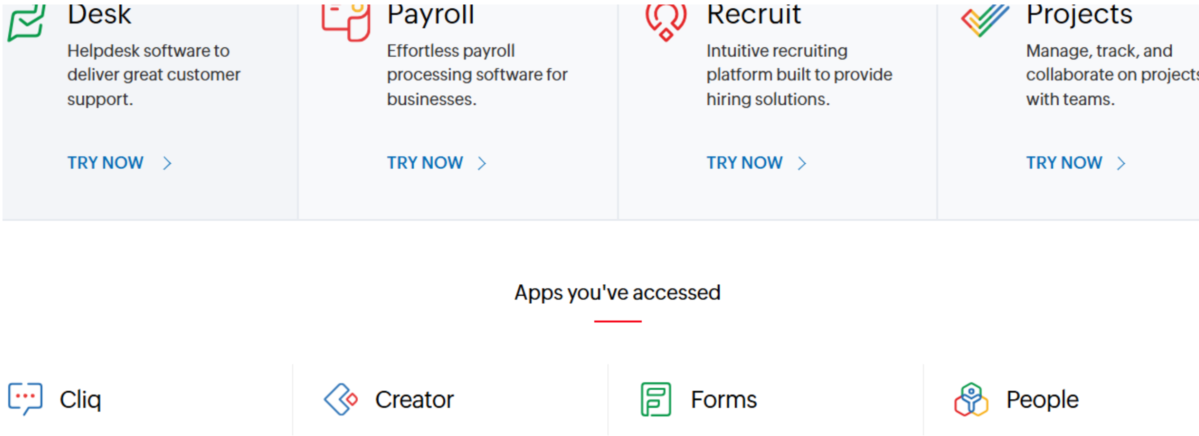
Aim:
To develop a cloud-based Mark Sheet Processing System using Zoho Creator.
Steps:

1. Create a new application.
2. Create a Student Mark Sheet form.
3. Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.
4. Use formulas to calculate Total, Average, and Percentage.
5. Save the records and verify the generated mark sheet.

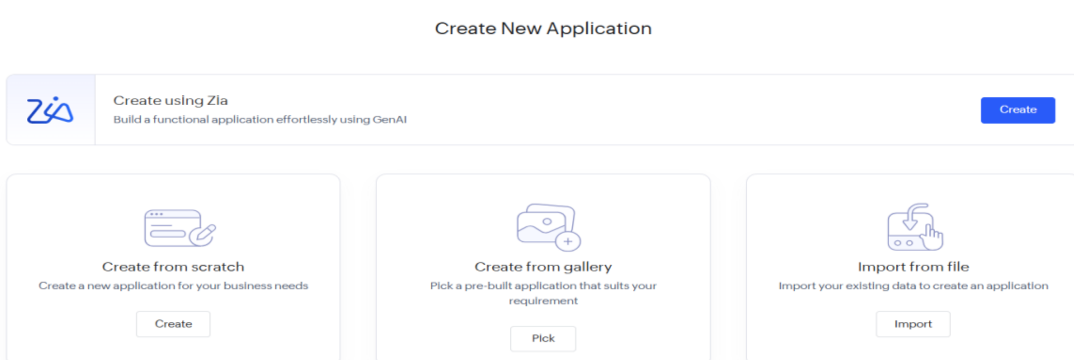
Implemented
Step 1: Go to zoho.com



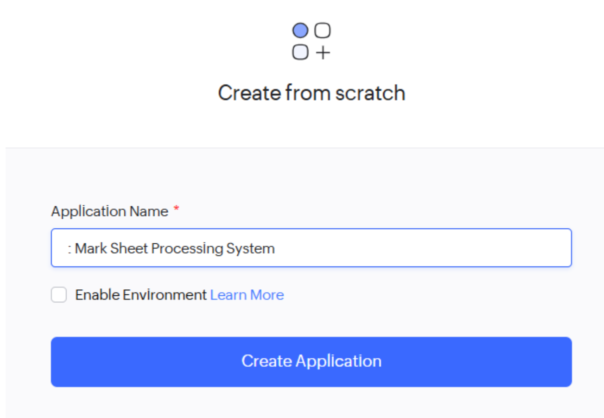
STEP 2: Login to your Zoho account and open Zoho Creator



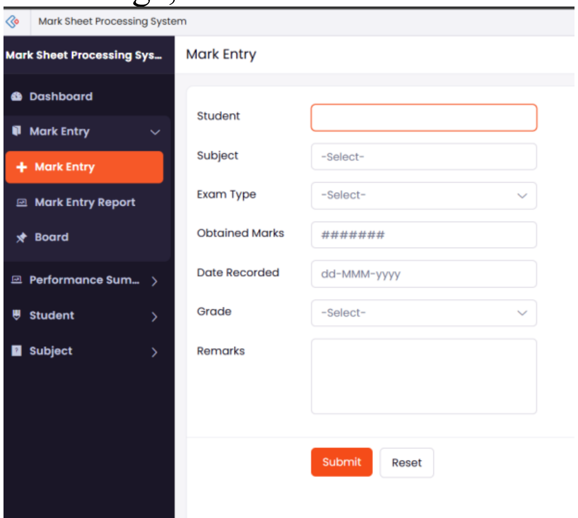
STEP 3: Click Create New Application and select Create from Scratch

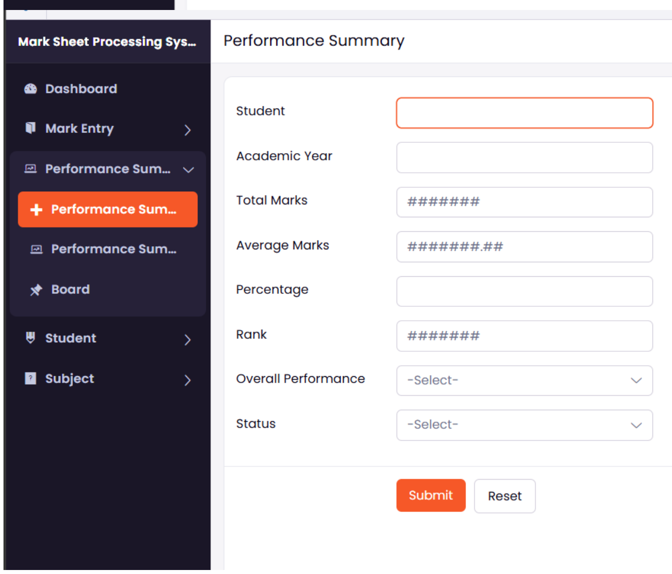


STEP 4: Enter Mark Sheet Processing System as the application name and click Create Application**.**

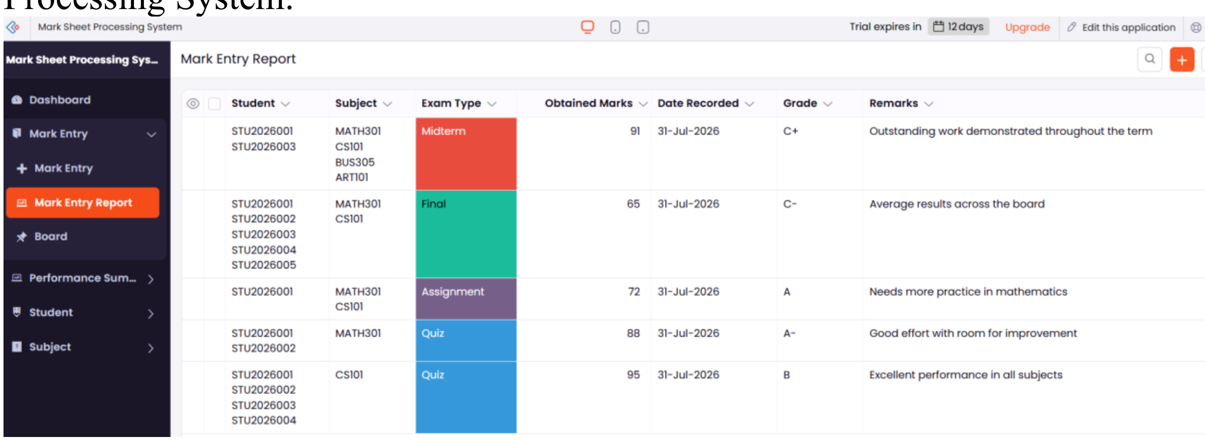


Step 5: Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.

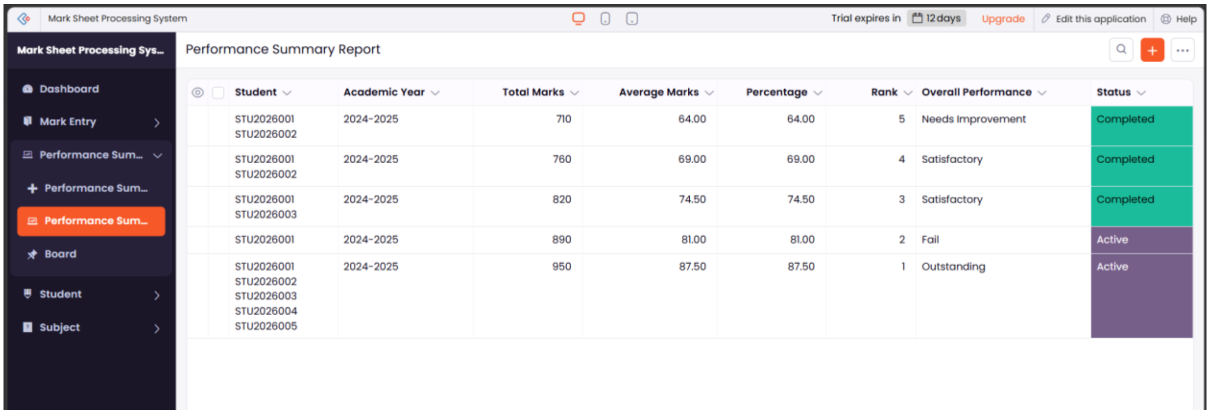




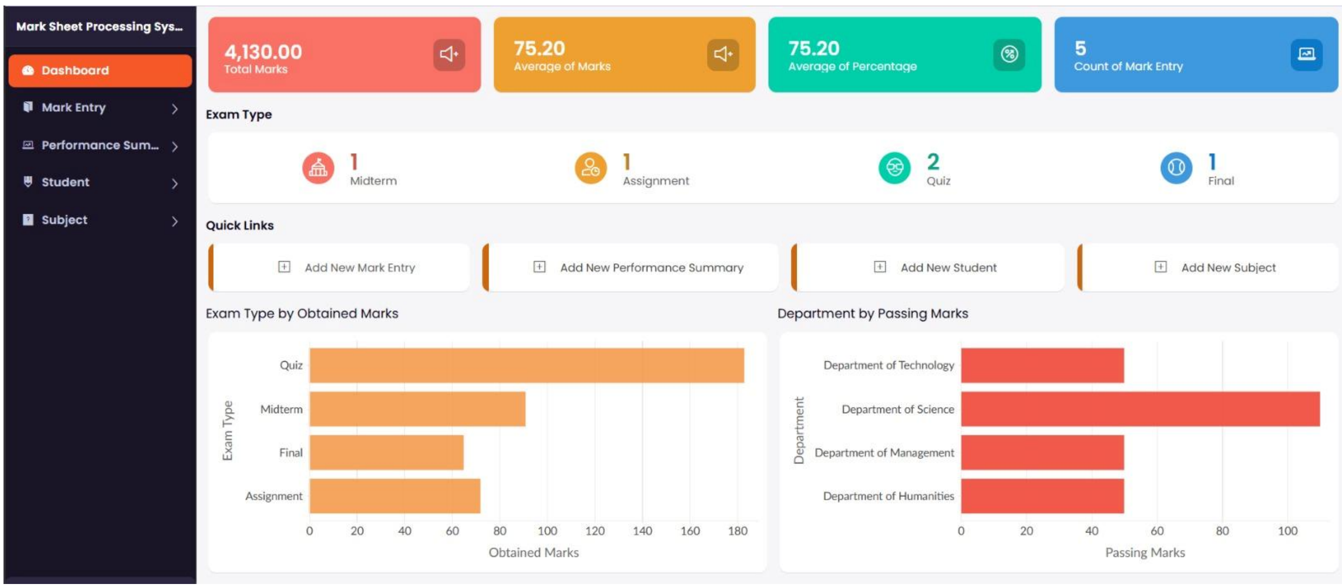
STEP 6: Save the form, enter sample booking records, and view them in the Mark Sheet Processing System.



Student	Subject	Exam Type	Obtained Marks	Date Recorded	Grade	Remarks
STU2026001 STU2026003	MATH301 CS101 BUS305 ART101	Midterm	91	31-Jul-2026	C+	Outstanding work demonstrated throughout the term
STU2026001 STU2026002 STU2026003 STU2026004 STU2026005	MATH301 CS101	Final	65	31-Jul-2026	C-	Average results across the board
STU2026001	MATH301 CS101	Assignment	72	31-Jul-2026	A	Needs more practice in mathematics
STU2026001 STU2026002	MATH301	Quiz	88	31-Jul-2026	A-	Good effort with room for improvement
STU2026001 STU2026002 STU2026003 STU2026004	CS101	Quiz	95	31-Jul-2026	B	Excellent performance in all subjects



Step 7:The Software has Been Created:



Result: The Mark Sheet Processing System was successfully created with student details, subject marks, average, percentage, rank, and overall performance. The application demonstrated SaaS-based online mark processing.

3. Install VirtualBox with different flavours of Linux or Windows OS on top of Windows 10 using custom installation.

Aim

To install VirtualBox and create a virtual machine with a different Linux or Windows operating system on the existing Windows environment.

Short Procedure

1. Download and install Oracle VirtualBox on the Windows system.
2. Download the ISO file of the required Linux/Windows OS.
3. Open VirtualBox and select Create New Virtual Machine.
4. Enter the VM name and select the downloaded ISO file.
5. Allocate suitable RAM, CPU cores, and virtual hard-disk space.
6. Start the virtual machine and follow the OS installation instructions.
7. Complete the installation and restart the virtual machine.
8. Verify that the newly installed OS runs successfully inside the virtual environment.

FIG : CREATING VM AND PROVIDING ISO FILE

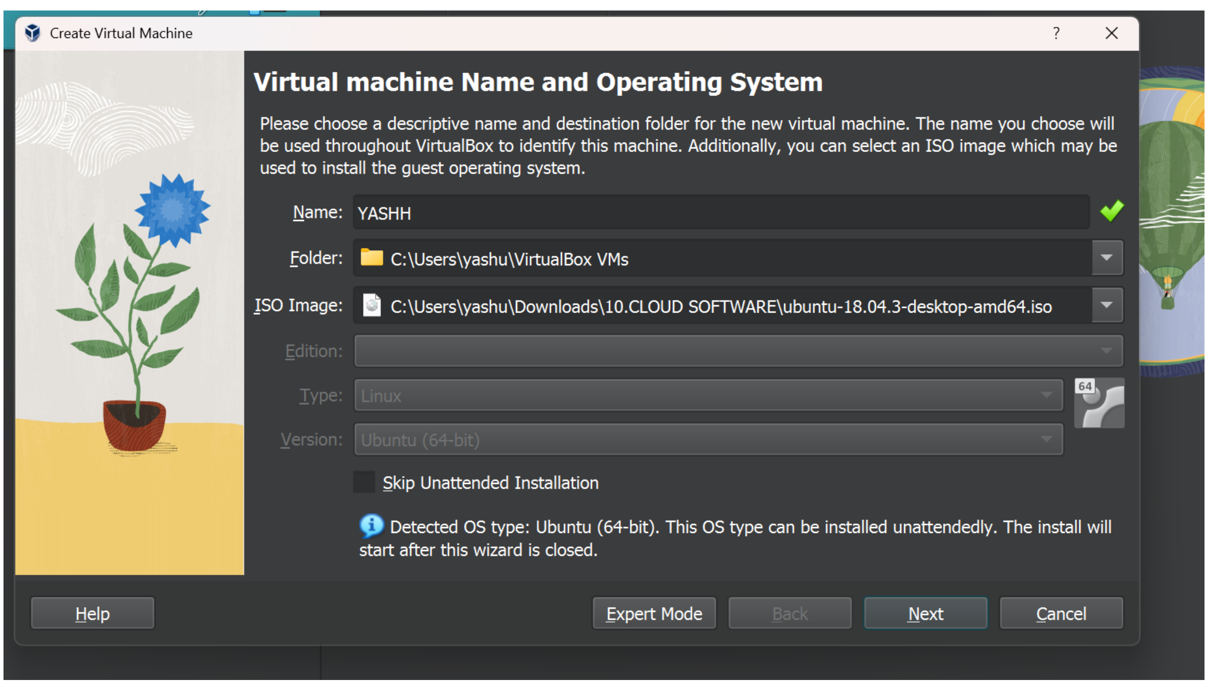


FIG :PROVIDING CREDENTIALS FOR THE VM

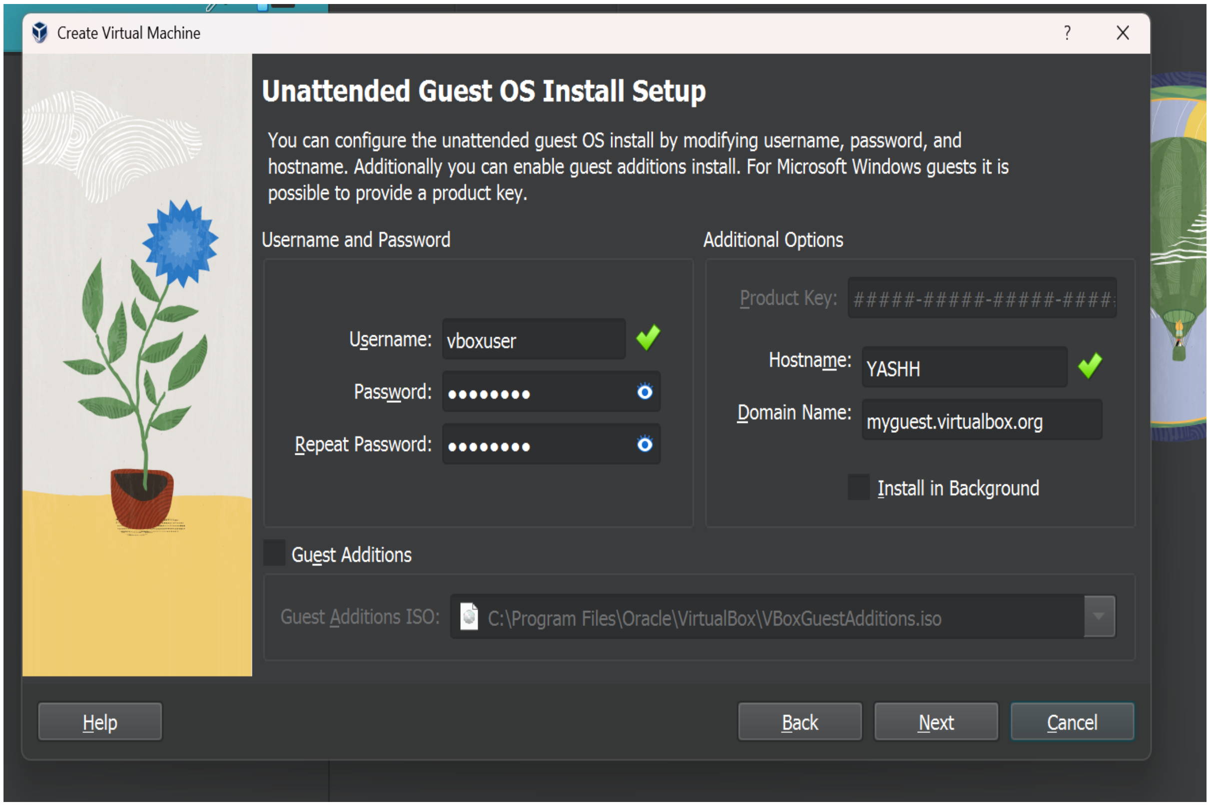


FIG : CREATING VIRTUAL HARD DISK WITH REQUIRED CONFIGURATION

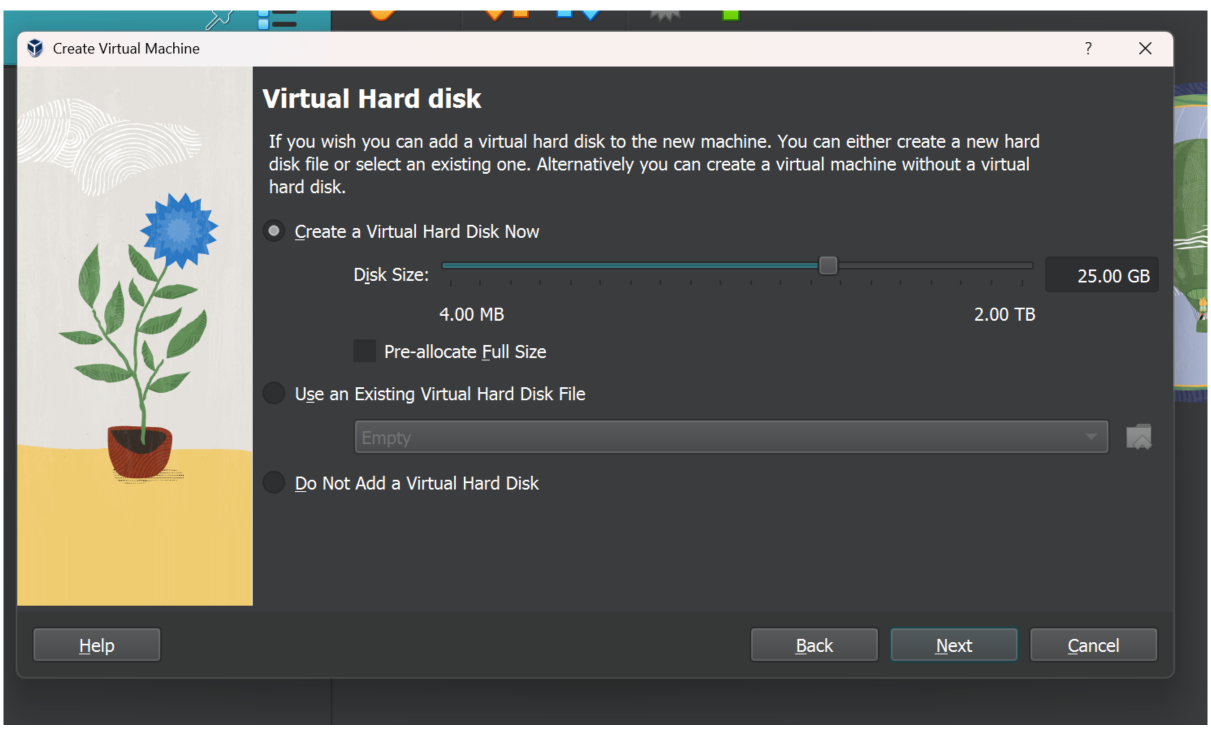


FIG : CONFIGURING THE VM

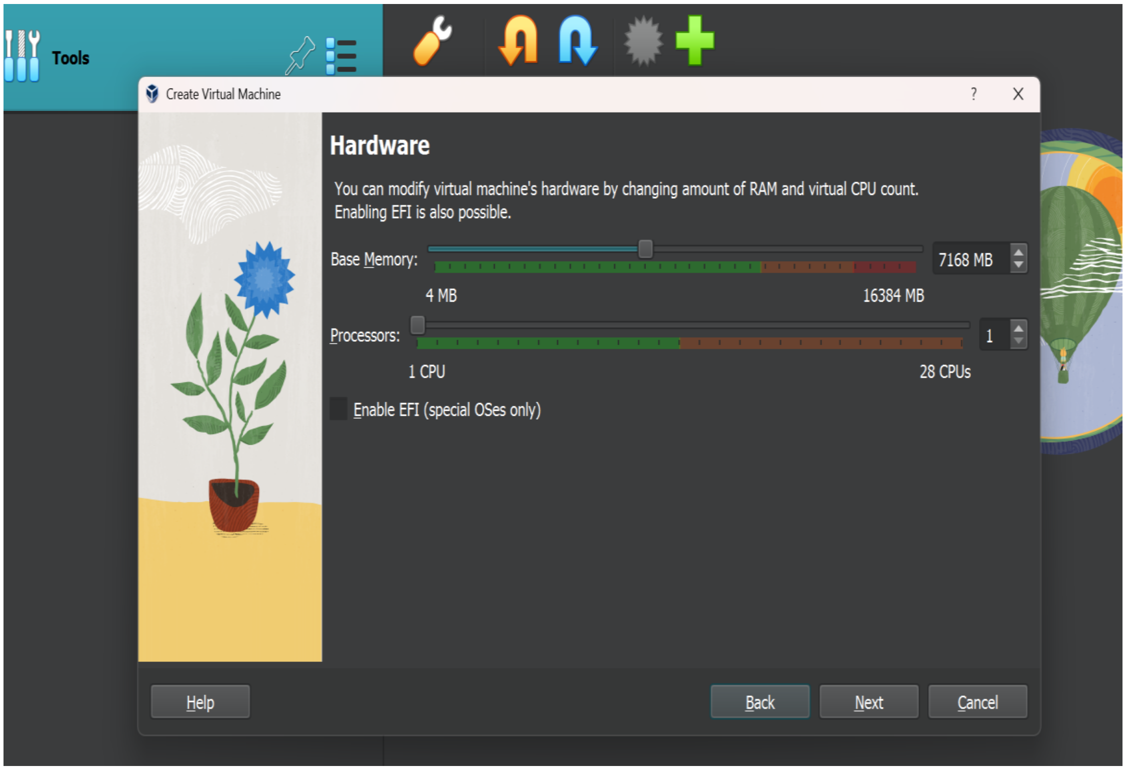


FIG : LAST STEP OF VM

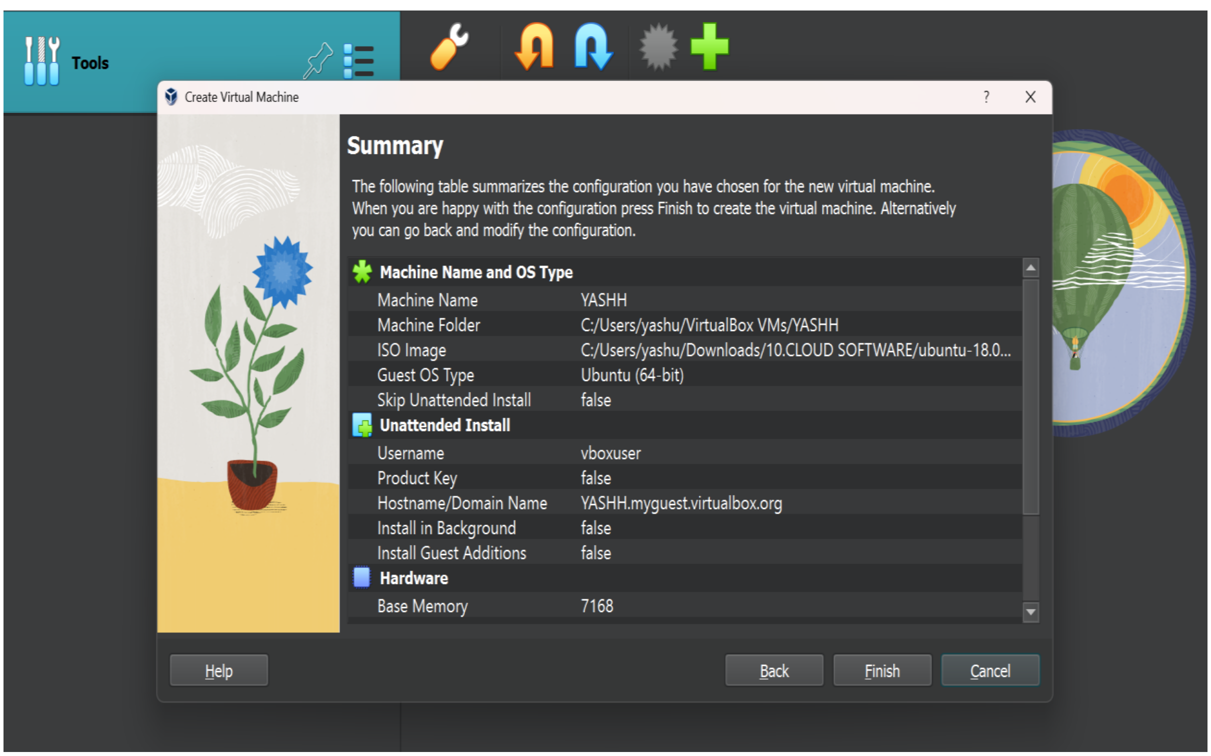
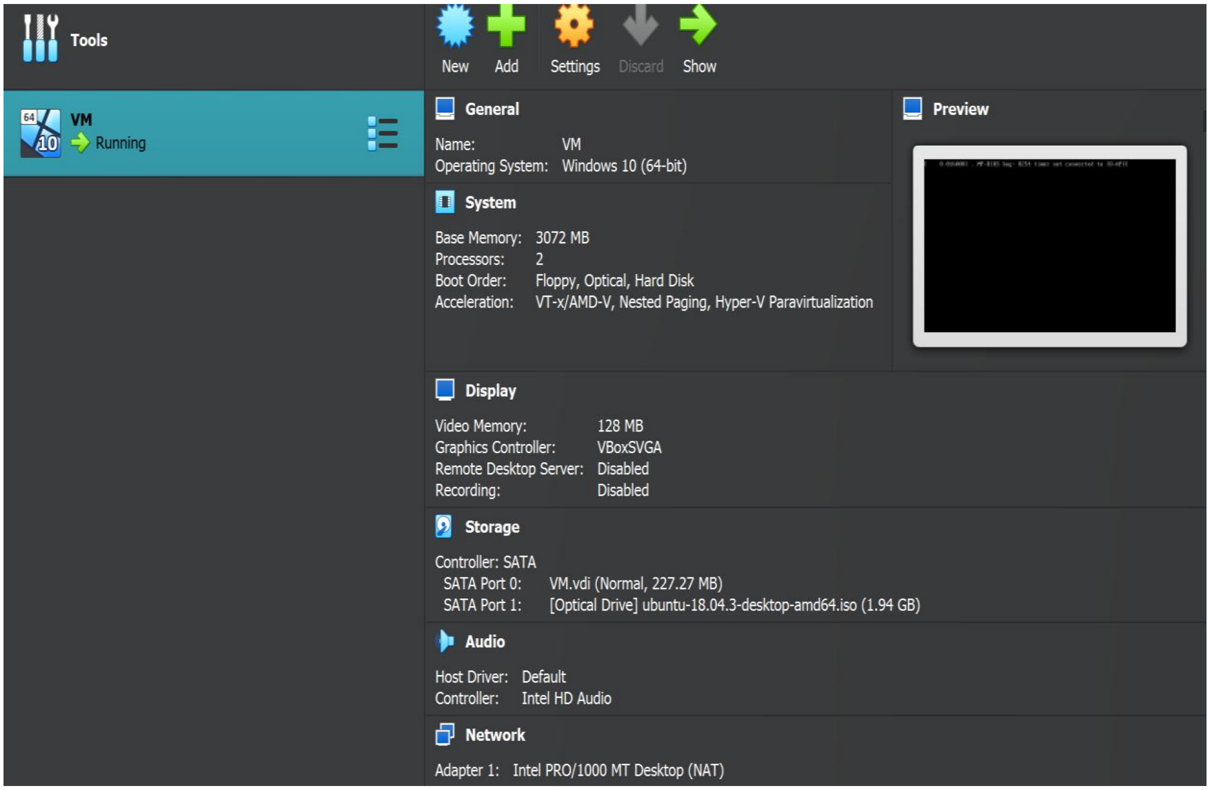


FIG : INSTALLATION OF VM IS FINISHED



Result: VirtualBox was successfully installed on Windows 10, and different Linux/Windows operating systems were installed as virtual machines using custom installation. The VMs were successfully configured and executed.

4.To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

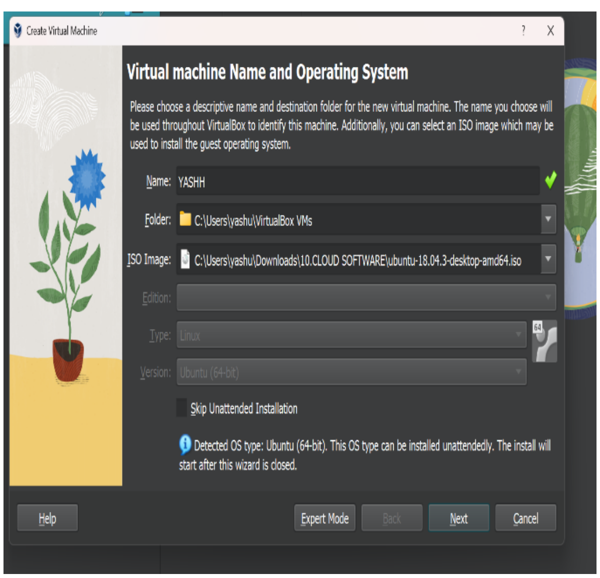
Aim :

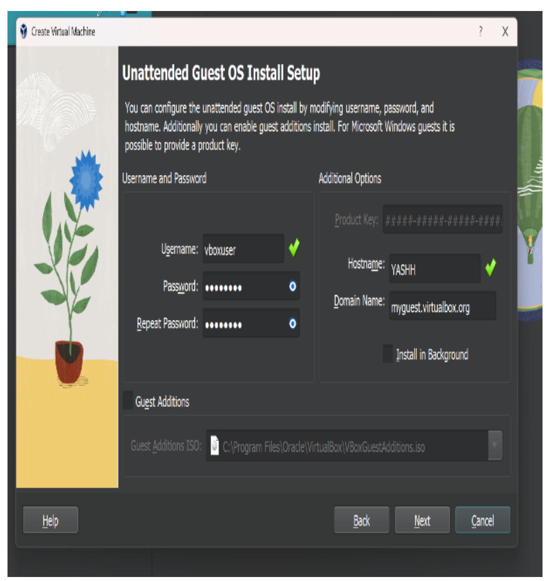
To run virtual machines with different configurations and determine the maximum number of virtual machines that can be utilized simultaneously on a host system.

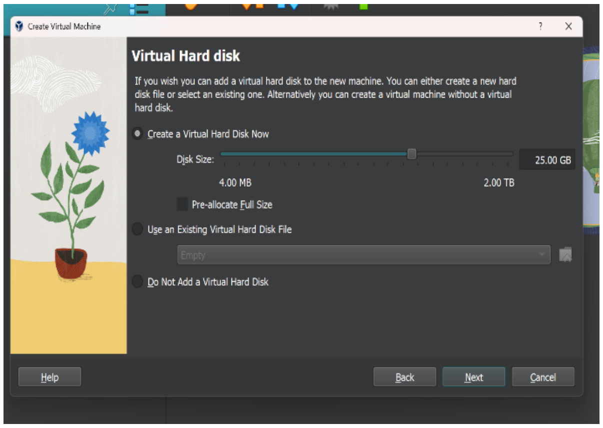
Short Procedure :

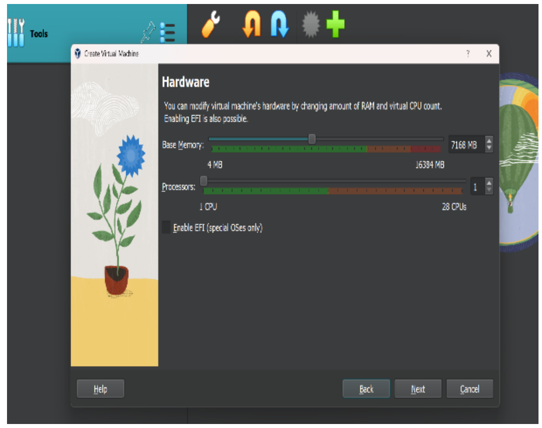
1. Open the virtualization software such as VirtualBox.
2. Create a virtual machine with a selected configuration (CPU, RAM, and storage).
3. Install the required operating system in the virtual machine.
4. Start the virtual machine and verify that it is running properly.
5. Create additional virtual machines with different configurations.
6. Start multiple VMs simultaneously and monitor CPU, RAM, and storage utilization.
7. Gradually increase the number of running VMs until the host system reaches its resource limit.
8. Record the maximum number of VMs that can run simultaneously without significant performance degradation.
9. Compare the results for different VM configurations and note the resource utilization.

FIG : PROCESS OF CREATING A VM Like above two more VMs are installed









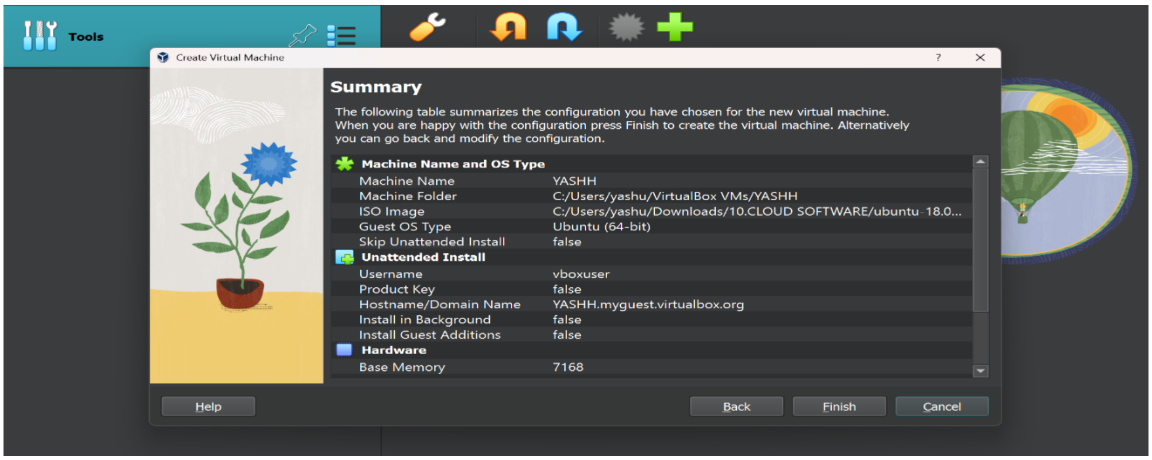


FIG : INSTALLATION OF VM YASHU

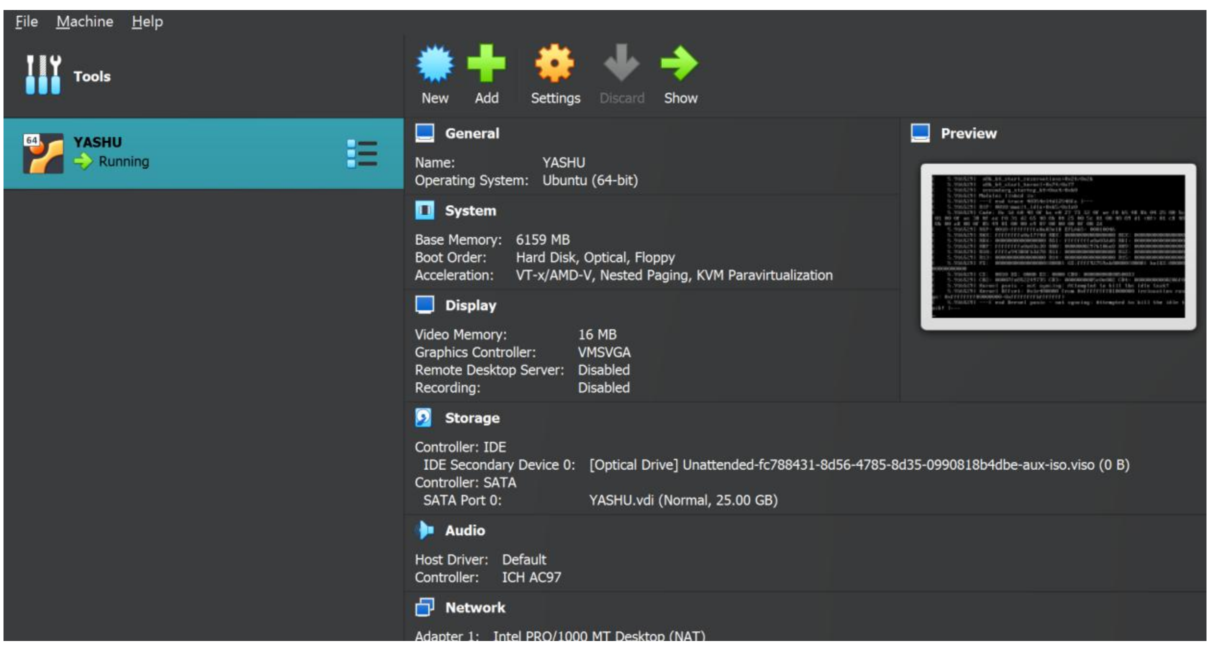


FIG : ACTIVITY STATUS OF VM YASHU

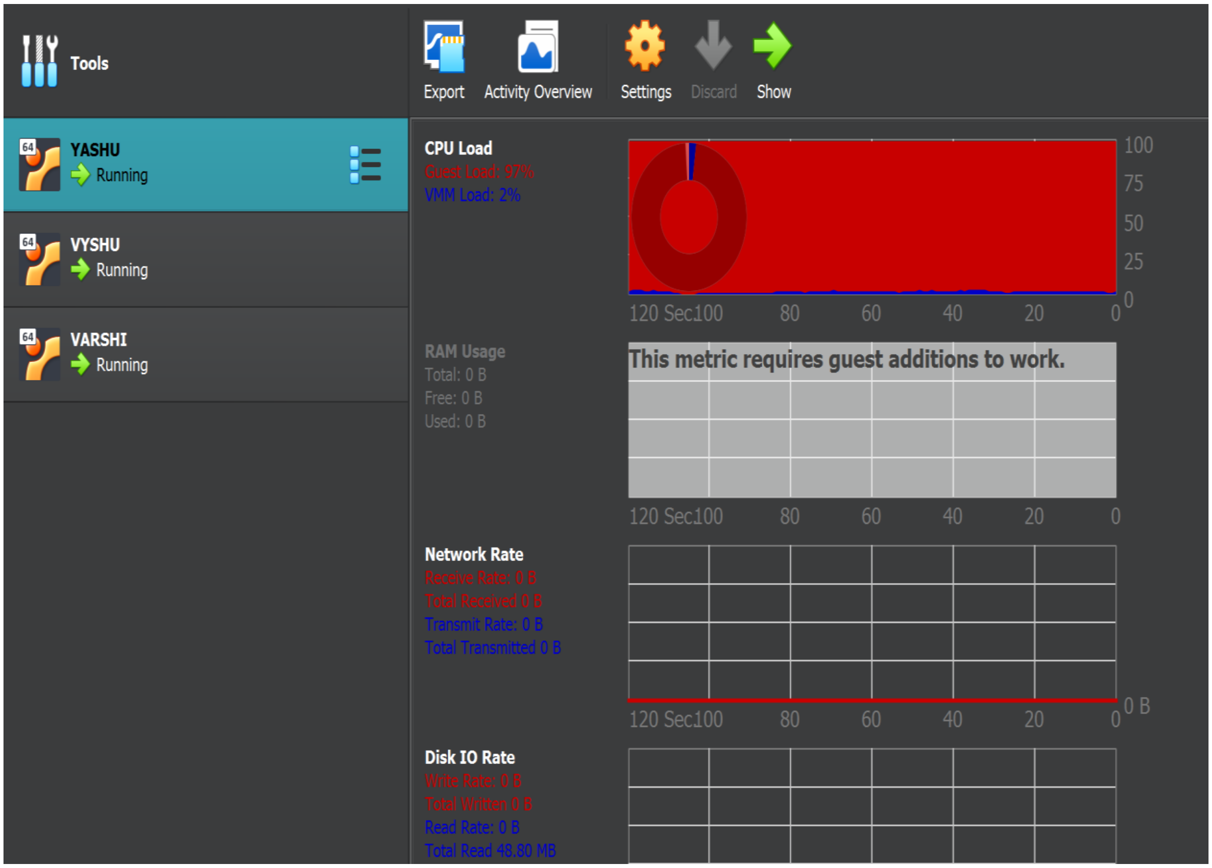


FIG : INSTALLATION OF VM VYSHU

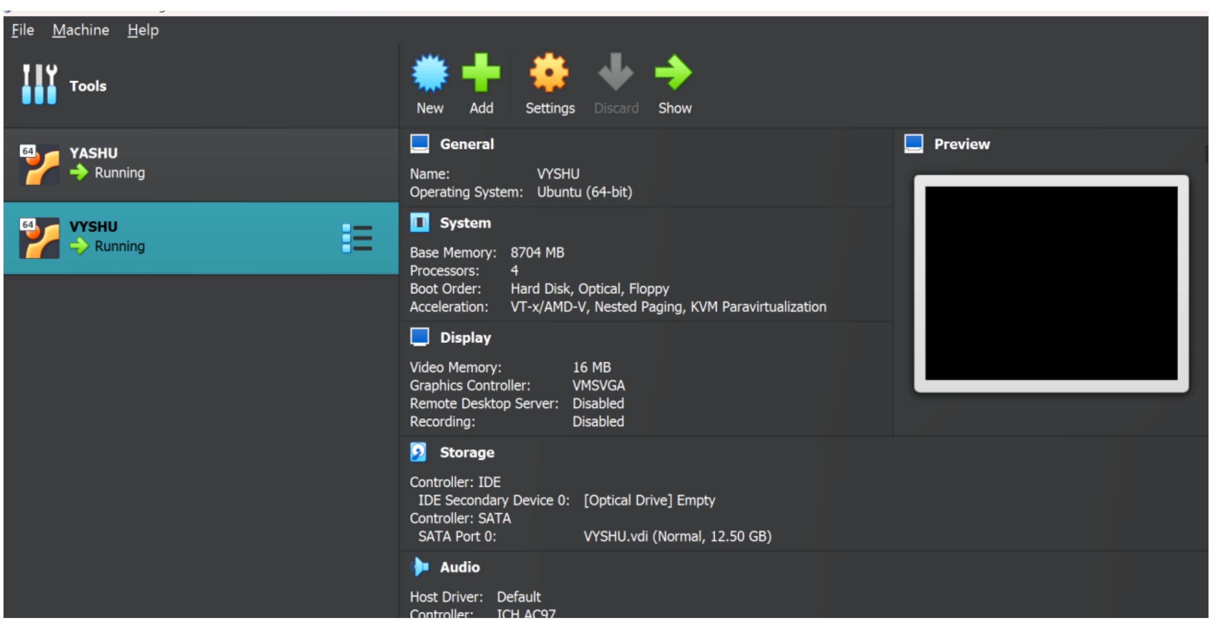


FIG : ACTIVITY STATUS OF VM VYSHU

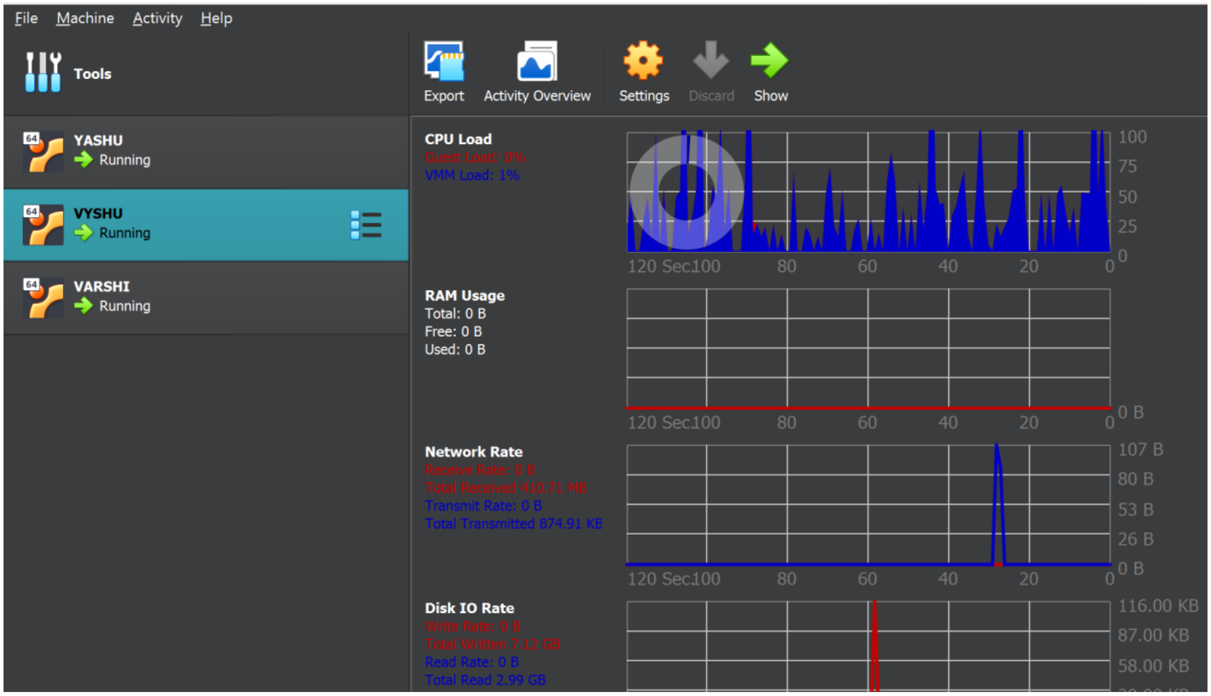


FIG : INSTALLATION OF VM VARSHI

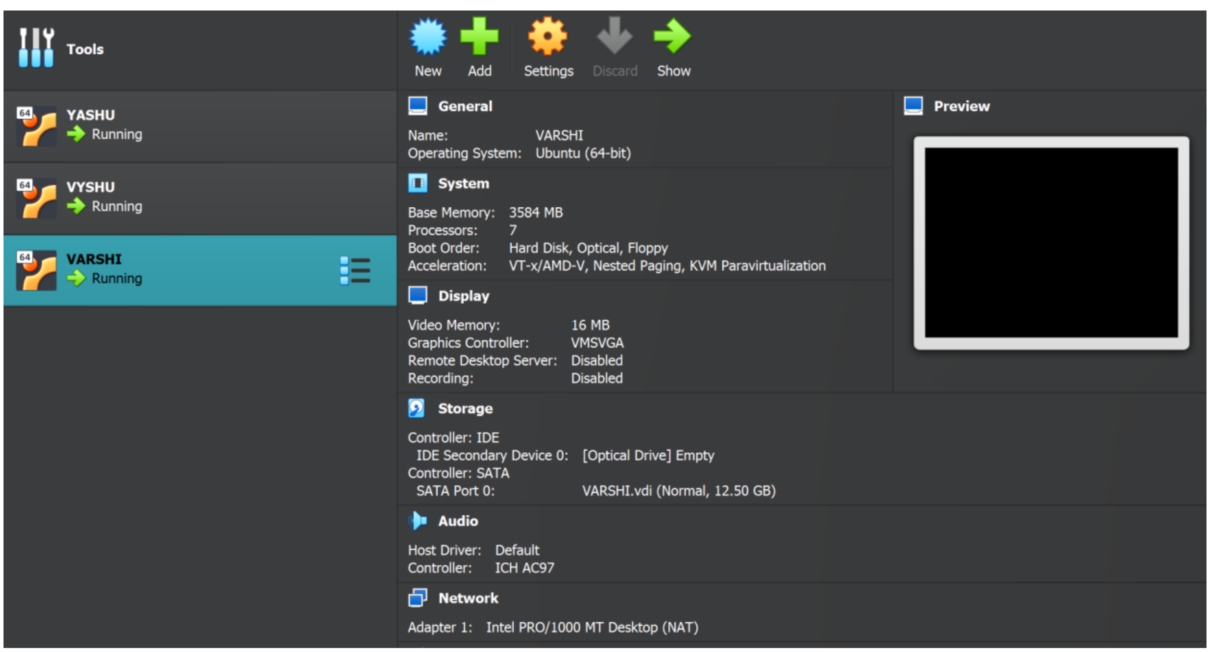
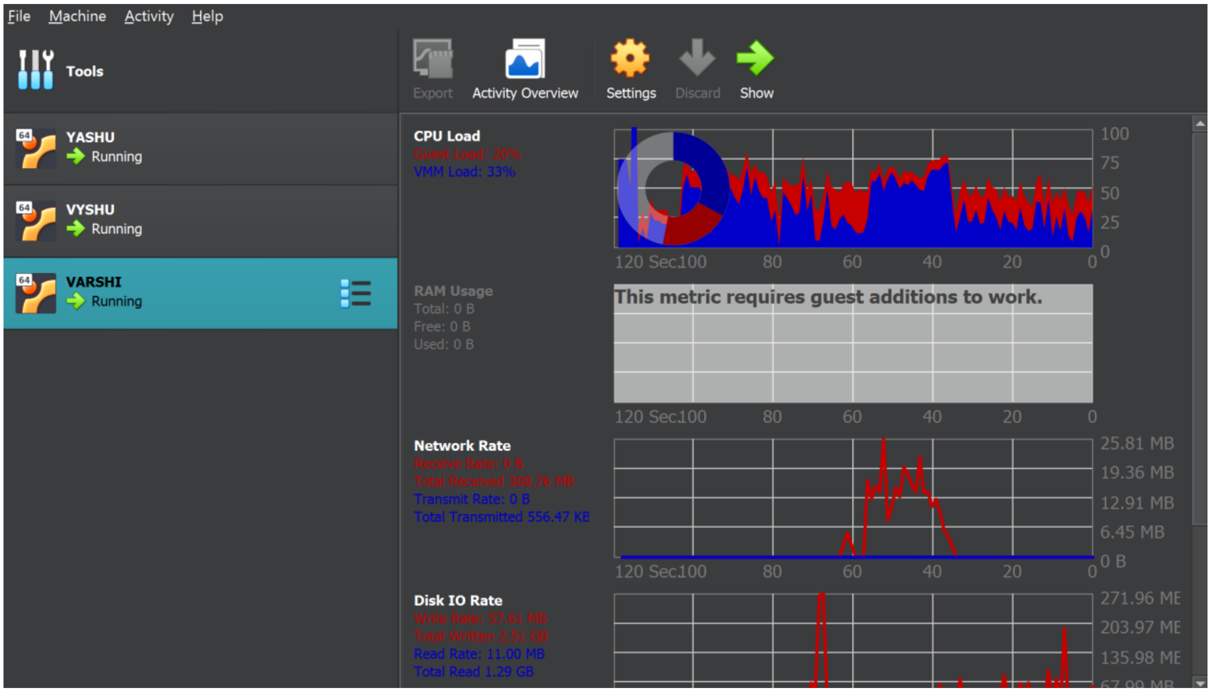


FIG : ACTIVITY STATUS OF VM VARSHI



Result: Virtual machines with different CPU, memory, and storage configurations were successfully created and executed simultaneously. The number of VMs that could be utilized at a particular time was determined by monitoring the host system's CPU, RAM, and disk utilization.